TITLE: Evolved change in orofacial motor circuits under selection

Introduction:

The Russian farm-fox experiment, since 1959, has allowed researchers to study the intergenerational effects of artificial selection on behavior, serving as a long-running longitudinal model of domestication (Dugatkin, 2018). The experimental set-up consists of two experimental breeding lines demonstrating either tame or aggressive behavioral phenotypes and a third control line in which there has been no selection for behavior. In its original conception, Beylyaev hypothesized that selection for specific behavioral traits in domestication would lead to a type of destabilizing selection that would have a diverse range of systemic effects during growth and development, leading to changes in the adult phenotypic expression across physiological systems (Beylyaev, 1979). The three breeding lines in this experiment are maintained through artificial selection for either affiliative or aggressive phenotypes over the first 8 months of life in the farm-foxes with handlers testing approach and approach-avoidance behaviors in young foxes until the final selection period at 8 months of age (Trut et al., 2016). The behavioral profiles we see in the current fox populations are thought to be as a result of this controlled breeding program and are largely genetically controlled. At this moment, this is the longest-running model we have for a controlled experiment in domestication in any species and the closest model for how selection for prosocial behaviors could lead to the types of selection that scaffold humans through enculturation processes in the world at large (Losey, 2022).

Understanding this model might reveal some insight into the processes of evolved change in the circuits of social communication during the long process of human evolution. The sociocommunicative facilities of the farm-foxes within this are central to the selective breeding program developed in this model of domestication (Kukekova et al., 2008). When separating young foxes into experimental groups, orofacial motor behaviors like licking, grinning and vocal behaviors like growling or barking are defining characteristics that allow for well-defined separation into an experimental group that will contribute to the continuation of the breeding line (Trut, 1999). Tame foxes, which exhibit affiliative or pro-social behaviors like mouth-hand-holding, allowing handler contact, whining, and tail wagging, are contrasted with aggressive foxes that show approach-avoidant and aggressive behaviors like pinned ears, tail raising, biting, barking, and maintaining peripheral distance from the handlers during interactions (Trut, 1999; Kukakova et al., 2008). Communication within these interactions during caretaking activities largely relies on social communication, such as facial expression, vocalization, and upper body posture, by the foxes. These behaviors, or analogs of them, are present currently across human populations and are abstracted to emotional behaviors that underlie more complex relationships like friendship or rivalry (Koelsch et al., 2015). While the literature contains a great body of information related to these circuits (Jürgens, 1970; Burrows, 2008; Székely et al., 1993; Jürgens, 2009; Cerkevich et al, 2022), no one study has documented the effects of selection on these circuits within a controlled model where orofacial behaviors are key circuits under selection.

Merit of project:

For this project, we aim to study structural components of the connectivity and organization of the facial nucleus and hypoglossal nucleus as a proxy for evolved change in these orofacial motor circuits in a domestication model to determine if selection for different communicative capacities is a strong enough driver to change the composition of these nuclei at the level of the brainstem and if selection on sociocommunicative faculties can lead to increased cortical connectivity to these brainstem areas in highly conserved circuits for orofacial motor behaviors. Previous studies have highlighted the need to understand the issue of convergence of cortical and brainstem-level circuits for complex behaviors like orofacial motor control, as neither exists structurally or functionally in isolation (Moore et al., 2011). Understanding the effects of selection on these circuits might provide some insights into natural processes of enculturation that emphasize social behaviors in human populations given our unique position within the context of communicative complexities like language and culture which both rely on social communication and social cohesion. Ideally, demonstrating that these systems can evolve in a model where an organisms are undergoing selection on social behaviors from humans helps us to understand the importance of drivers of human social bonds and social cohesion, like facial expression and vocal communication. This study provides a general framework for how selection could have led to the emergence of specialization for both the production and perception of finely-tuned orofacial movements.

Neural Circuitry Governing Orofacial Movements:

Control of orofacial movements varies across species, but these circuits have some conserved characteristics across mammalian species. (See Figures IV and V in the appendix for a synthesis of connectivity patterns of the cortical circuit and brainstem level circuit in a subset of mammals.) In most mammals, control of facial expression involves circuits composed of both cortical and brainstem-level components. In primates and humans, the facial nucleus receives parallel cortical input from the primary motor cortex and anterior cingulate motor cortex, both of which contain facial motor homunculi (Strick et al, 1978; Strick et al, 1998; Belyk et al., 2021a; Belyk et al., 2021b; Cerkevich et al., 2022; Zhang & Ghazanfar, 2022). In this circuit, the muscles of the upper face receive bilateral innervation, while the muscles of the lower face receive unilateral input from the contralateral cortex (Müri, 2016). In cats and domesticated dogs, there are three orofacial representations of the orofacial muscles that innervate the facial nucleus: area C, which is located in the anterior part of the coronal gyrus; area P, which consists of the lateral wall of the presylvian sulcus; and area M, located on the ventral bank of the cruciate sulcus (Hiraba et al., 1993; Milojevic & Hast, 1964). The patterns of cortical innervation of the facial nucleus in cats and dogs consist of bilateral projections from areas C and P and unilateral and mixed contralateral and ipsilateral projections from area M (Schmitt & Gacek, 1986; Felleman et al., 1983; Hisa et al, 1982). Coordination of these cortical regions, largely considered to be orofacial motor regions, contributes to the control of the muscles of facial expression.

Neural circuits for motor control for articulation, largely dictated by the movement of the tongue, also consist of both cortical and brainstem-level components (Barnard, 1940). In humans and other primates, the hypoglossal nucleus is largely bilaterally controlled by multiple descending projections for premotor and primary motor regions, with the functional driving force of most tongue movements being driven by contralateral input in these circuits (Morecraft et al, 2014; Tram et al, 2023; Gerek & Binar, 2015). In cats and domesticated dogs, the hypoglossal nucleus receives bilateral input from areas C and P, as well as the prorean gyrus (Hiraba et al., 2013; Hiraba et al., 2004; Hiraba et al., 2005). Like facial motor control, tongue motor control appears to have modular characteristics but not an entirely structurally or functionally lateralized pattern. Unilateral lesions to area P, instead of producing horizontal deviations across the midline, produce errors in tongue protrusion and retraction in cats (Hiraba et al., 2009). Connectivity patterns that consist of combinations of unilateral and bilateral input contribute to the control of the muscles for articulation of the tongue for vocal motor behaviors.

Broader Implications:

Complex communicative systems in this domestication model serve as a means to understand how selection for sociocommunicative behaviors can affect the structures of the central nervous system within foxes. However, these same systems also comprise the dynamics of social environments in the cultural systems of humans of which are considered to be under selection for prosocial behaviors within the context of human evolution. First noted in the early work of Darwin (Darwin, 1868) who discussed the similarities between modern humans and domesticated animals, this framing has been developed (Hare et al., 2021; Wrangham, 2018; Wrangham 2019) and is currently known as the human self-domestication hypothesis (Hare et al., 2013). While Wrangham describes human populations under negative selection for reactive aggression, Hare has taken this one step further to suggest that cooperation and positive selection for prosocial behaviors like friendliness are the primary drivers of current human evolutionary change. Sánchez‑Villagra & Van Schaik, 2019 further critique and refine the views presented by Wrangham and Hare, but expand into niches outside of social behaviors to consider other behavioral and physiological changes underlying the human self domestication hypothesis. This project was developed to address issues of selection on orofacial motor behaviors that could lead to changes in neuroanatomical systems when sociocommunicative systems and prosocial behaviors are under selection within a domestication model.

Aim Summaries:
Our first aim will look at evolved changes in the brainstem circuits for communication, where we apply a structural analysis to the facial nucleus and hypoglossal nucleus across the experimental and control groups by quantifying the neuropil fraction of the subregions of these nuclei, which provide quantitative measures of the neuropil including axons, synapses and dendrites. We will also determine the distribution of cell size across these subregions and across groups to identify whether changes to cell size and cell number have occurred in response to selection. We will then determine the overall volume of these nuclei to determine if the overall size of the nucleus has changed under artificial selection. The goal of Aim I is to determine if there has been an evolved change in these highly conserved circuits that control the muscles for orofacial motor behaviors and receive their driving input from multiple cortical regions.

Our second aim will utilize MRI and DTI tractography to investigate whether the cortical innervation to key nuclei involved in orofacial motor behaviors is increased, indicating a greater degree of cortical control over these behaviors. Specifically, we are interested in examining whether this increase in cortical control is associated with the selection for sociocommunicative behaviors. We aim to compare the levels of cortical involvement in orofacial motor functions between tame and aggressive foxes, with a particular focus on their facial expression and vocal-motor behaviors. By doing so, we hope to better understand how selection pressures associated with domestication might influence brain organization and control of these behaviors. Ultimately, our goal is to explore the role of selection in driving neuroanatomical changes, particularly in the context of domestication, by characterizing the cortical and brainstem circuits that regulate orofacial motor behaviors at multiple levels of neural organization. This approach will provide insights into how evolved changes in neural circuits contribute to sociocommunicative behavior in foxes and other mammals.

Aim 1: Evolved changes in brainstem circuits for communication in the Russian farm-foxes in response to artificial selection

Aim 1 Goal: To determine whether the brainstem circuits influencing aspects of social communication, specifically the facial motor nucleus and hypoglossal nucleus, exhibit evolved changes as an adaptation to artificial selection for affiliative or aggressive behavioral phenotypes in a domestication model.

Aim 1 Hypotheses:
H1: The neuropil fraction of the facial nucleus will differ across fox populations.
H0: The neuropil fraction of the facial nucleus will not differ across fox populations.
H2: The neuropil fraction of the hypoglossal nucleus will differ across fox populations.
H0: The neuropil fraction of the hypoglossal nucleus will not differ across fox populations.
H3: Cell size in the facial nucleus will differ across fox populations.
H0: Cell size in the facial nucleus will not differ across fox populations.
H4: Cell size in the hypoglossal nucleus will differ across fox populations.
H0: Cell size in the hypoglossal nucleus will not differ across fox populations.
H5: The volume of the facial nucleus will differ across fox populations.
H0: The volume of the facial nucleus will not differ across fox populations.
H6: The volume of the hypoglossal nucleus will differ across fox populations.
H0: The volume of the hypoglossal nucleus will not differ across fox populations.

Aim 1 Justification and Background:

TITLE: Evolved change in orofacial motor circuits under selection

The facial nucleus and hypoglossal nucleus are structures within the pontomedullary brainstem that influence the control of orofacial movements in vertebrate species. (For innervations patterns of the facial nucleus in chimpanzees and cats see Figure X and for innervations patterns of the tongue in canines and humans see Figure II.) These regions, composed of premotor neurons, contribute to fine motor control of the upper and lower face and movements of the tongue (Li et al., 1993). The morphology and composition of these nuclei have been associated with different aspects of social group formation and social communication in vertebrate species (Sherwood et al., 2005; Dobson & Sherwood, 2011). Larger numbers of premotor neurons in the facial nucleus have been associated with complex facial movements in great apes and humans, while the volume of both the hypoglossal nucleus and facial nucleus has been associated with a species degree of facial mobility and social group size in nonhuman primates species (Sherwood et al., 2005).

The circuits of which the facial nucleus and hypoglossal nucleus are part of a stable circuit and thought to be a part of a highly conserved brainstem network in vertebrate species that controls movements of the orofacial region for communication and feeding, but there is evidence that suggests that there are notable functional influences on the growth trajectories of these nuclei during development (Dobson & Sherwood, 2011). In nonhuman primates, the hypoglossal nucleus, facial nucleus, and closely associated trigeminal nucleus all have independent growth trajectories maintained during periods of cortical development. These nuclei scale in response to feedback from developing cortical regions. While there is a degree of functional plasticity during development, the growth of the facial nucleus and hypoglossal nucleus can be expected to scale phylogenetically, meaning these nuclei decouple from projected volumes based on brain volume, and when under selection, the volume of these nuclei will also decouple from phylogenetic patterns (Sherwood et al., 2005).

When volume increases are observed in these nuclei, it is often associated with increased white matter within the individual nucleus (Sherwood, 2003; Sherwood et al., 2005). White matter increases notably signal increased connectivity between coupled nuclei within a localized circuit or increased connectivity with higher-level brain regions like the cortex (Palomero-Gallagher & Zilles, 2019). These changes can be accompanied by additional constraints within these developing nuclei. In the facial nucleus, increased white matter volume is also paired with increased numbers of premotor neurons within the nucleus within studies of primate groups. In the hypoglossal nucleus, no accompanying somata changes have been observed with white matter increases within the nucleus within similar primate species (Sherwood et al., 2005). This suggests that these regions may demonstrate differential aspects of adaptation to aspects of selective pressures due to the dynamics of each group's social system.

To better understand how selection shapes these systems within social environments, it becomes necessary to understand how other organisms in complex social environments have responded to selective pressures in increasingly complex systems. Animals under domestication offer a window into this process, whereby nonhuman animal social environments meet artificial selection implemented within human-centered environments for functional or breeding purposes (Kruska & Stephan, 1973; Kruska, 2005). The functional and structural response to domestication varies by species, and this can include processes like changes to brain scaling and brain reorganization (Kruska, 2005). Scaling is markedly different across domesticated species (See Figure I for examples). While in most domesticated species, the telencephalon is the most reduced structure within the central nervous system, the medulla and mesencephalon are the least reduced structures within this system. This is true for species like the domesticated dog, which was the basis of the model for the farm-fox domestication experiment (Kruska & Stephan, 1973). In wild and lab-reared populations, white matter and grey matter typically scale allometrically within mammalian species (Zhang et al., 2000). Increased white matter in these models scales at a much higher rate for long-range projections than for local circuits in mammalian species (Zhang et al., 2000). Thus, we might infer that volumetric changes even in domesticated species might be largely dictated by the degrees of connectivity of these long-range projections.

Tame and aggressive farm-foxes in this model express marked behavioral differences in orofacial gestures, and we believe that these behavioral profiles are observable and can be differentiated at the cellular level. Tame foxes vocalize more in response to unfamiliar humans within their enclosures but generally allow approach behaviors and handler contact (Gogoleva et al., 2011; Volodin et al., 2008; Kukakova et al., 2008). They also cackle and pant instead of barking, which are vocalizations that are unique to their behavioral group (Gogoleva et al., 2008). Aggressive foxes bark and vocalize, in general, more frequently than either the tame or conventional fox groups, but exhibit more aggressive postural behaviors (Kukakova et al., 2008; Gogoleva et al., 2008). No current study has documented the differences in facial gestures under this domestication model, but we were able to acquire the behavioral data from Kukekova et al, 2008. We separated orofacial behaviors from postural behaviors in this data set and performed a principal component analysis. Tame foxes in this analysis had distinct grouping for both of these behavioral traits compared to aggressive and conventional foxes. Additionally, we performed a covariance analysis of fox behaviors when handlers initiated touching, and tame foxes behaviors more positively covary with handler touch initiation than the aggressive and conventional groups. (See Figure VII Figure XI for these analyses). The orofacial movements of each group utilize muscle groups differentially, and we would expect a degree of plasticity in the circuits controlling these orofacial gestures, as selection can decouple scaling laws from phylogenetic patterns at the brainstem level as well as in cortical circuits. Volumetric and compositional differences could be observable at the cellular level.

To test the hypothesis that selection on sociocommunicative behavior can differentially alter the composition and structure of these brainstem-level nuclei, we have planned to perform a neuropil fraction and cell profile analysis as well as utilize a stereological probe to determine the volume of these nuclei from histological sections. The neuropil fraction provides measures of the neuropil, which can consist of white matter, synapses and dendrites within a given region of interest (Spocter et al, 2012). The neuropil fraction can serve as proxy for the interconnectedness of a group of cell bodies, in this case for the reception of long range connections from cortical and subcortical regions (Schlaug et al., 1993; Pannese, 1994; Sherwood et al., 2004). See Figure VI within the appendix for example images from this pipeline. Volumetric measures would provide a whole nucleus structural analysis (Haug, 1987). Additionally, we would measure cell size and determine if cell size differed in each nucleus across these fox groups under different selective pressures. Previous implementations of these methods can be found in Issa et al., 2019 and Spocter et al., 2012. We would anticipate that if vocal behaviors or facial gestures are under selection in this model, we would observe increased neuropil fraction within these nuclei, signaling increased long-range connections, presumably from cortical regions, to these nuclei. Larger cells are often associated with long-range connectivity and strong driving output, the most famous of which are the Betz cells in the motor cortex or Von Economo neurons within prefrontal regions (Goulas et al, 2018). We might also anticipate that cell size may change in response to these selective pressures, and we would anticipate that if facial gestures are under selection in this model, we might see an adaptive change for larger cell bodies within the facial nucleus to meet sufficient driving output from this nucleus to the muscles for facial expression.

Aim 1 Materials and Methods:

Foxes used in this study were provided by the Institute of Cytology and Genetics of the Russian Academy of Sciences in Novosibirsk, Russia, where the farm-fox experiment has been ongoing since 1959. Animal procedures followed standards for humane care and use of laboratory animals, adhering to Office of Laboratory Animal Welfare Assurance guidelines. Brainstem tissue from the left hemispheres of 30 male foxes (10 tame, 10 aggressive, and 10 conventional) aged 1.5 years was prepared and analyzed using previously described Nissl staining methods (Rogers Flattery et al., 2023). High-resolution digitized images of 40 µm sections were obtained for neuropil fraction analysis and cell profile analysis.

Regions of interest were identified using MRI images paired with slide-scanned Nissl sections and cross-referenced with comparative atlases of foxes, domesticated dogs and domesticated cats (Rogers-Flattery et al., 2023; Palazzi & Palazzi, 2011; Berman, 1968). Sampling focused on the entire vertical length of the facial motor nucleus and hypoglossal nucleus using MBF StereoInvestigator, with a sampling grid of 129x129 µm and a sampling size of 100x100 µm. Images were processed in ImageJ, scaled to 0.685 µm/pixel, and analyzed using a custom macro for binarization and gray value measurements. The neuropil fraction is derived from the relative amount of white space within the sampled images to the total number of pixels in the entire region of interest. Images were additionally processed through ImageJ to determine cell number and cell size within each nucleus. Statistical analysis, which will include ANOVA and Tuskey's HSD, will be performed to evaluate the data.

An ANOVA will be performed to determine whether there are significant differences in the means of the three farm-fox groups. In the ANOVA, we would look at between-group variance to determine variability due to differences in group means and within-group variance, which will look at variability within each group due to individual differences or random error. If $p < 0.05$, the null hypothesis would be rejected, and post hoc testing would be performed to determine which groups differ significantly. If $p \geq 0.05$, the null hypothesis wouldn't be rejected, and no differences would be observed across groups. These statistical measures would be applied across the neuropil fraction, volumetric analysis, and cell profile analysis within Aim 1.

Aim 1 Preliminary Data:
Aspects of the neuropil and cell profile pipeline can be found in the appendix. This pipeline was used in a previous analysis to assess differences in the tame, aggressive, and conventional foxes in a frontal region of the cortex, and examples of this analysis are provided for proof of concept in Figure VI.

Aim 1 Significance and Interpretation:
If there are positive results or differences across the fox populations, then it could be interpreted that selection on social behavior can influence the organization and structure of brainstem circuits for social communication. Understanding these dynamics would open up an area of research to better understand comparative aspects of the plasticity of highly conserved circuits in the brainstem in response to behavioral modifications in humans and other mammals.

If there are negative results or no differences across the fox populations, then it could be interpreted that selection on sociocommunicative circuits rely additionally on "higher-order" circuit-level changes in the cortex or subcortex. This might suggest that social behaviors are more reliant on the convergence of higher-level systems with brainstem-level circuits than specific organizational changes at these levels independently, which would emphasize the need for an integration of neuroanatomical work looking at cortical and brainstem level changes within the same study.

To clarify, if there are differences across the neuropil fraction than selection on social behaviors could be said to affect white matter connectivity in these regions which could imply higher level control for volitional movement. If there are differences in the cell profile across groups, it could be interpreted that driving input to these nuclei have evolved to accommodate differential driving inputs. With larger cells, it is possible that driving forces through the nucleus to the muscle is higher with computations performed at the cortical level and not at the level of the brainstem. With smaller cells, but perhaps more numerous cells, it would be implied that driving forces are smaller and perhaps greater computation occurs at the level of the brainstem. Additionally, these results would have to be taken within the context of overall volume changes to the nucleus itself to understand if these are independent changes, or if neuropil or cell size has decoupled from volumetric considerations from either the nucleus or the medulla itself.

Aim 2: Evolved change in the cortical connectivity of neural circuits for communication in the Russian farm-foxes in response to artificial selection

Aim 2 Goal: To determine whether neural circuits involving the facial nucleus and hypoglossal nucleus for social communication exhibit increased cortical connectivity and thus robusticity in response to artificial selection.

Aim 2 Hypotheses:
H1: Cortical connectivity to the facial nucleus will differ across fox populations.
H0: Cortical connectivity to the facial nucleus will not differ across fox populations.
H2: Cortical connectivity to the hypoglossal nucleus will differ across fox populations.
H0: Cortical connectivity to the hypoglossal nucleus will not differ across fox populations.
H3: Tame foxes will have increased cortical connectivity from frontal motor regions to the facial nucleus due to artificial selection on facial gestures.
H0: No farm-fox group will have increased cortical connectivity to the facial nucleus due to artificial selection on facial gestures.
H4: Tame foxes will have increased cortical connectivity from frontal motor regions to the hypoglossal nucleus due to prevalence of directed vocal behaviors.
H0: No farm-fox group will have increased cortical connectivity to the hypoglossal nucleus due to artificial selection on vocal communication

Aim 2 Justifications and Background:
Facial expression and vocal motor behaviors like oral articulation fall under a broader range of movements characterized as orofacial movements. In laboratory species, primates and humans, orofacial movements are largely influenced by both cortical and brainstem level circuits. For example summaries of the interconnectivity of their circuits see Figures IV and V within the appendix; we see from these figures that as brainstem networks become less robust, there are changes to the robustness of the cortical input to these regions which can be observed across species. Of these circuits, we will draw a particular focus on the cortical control of the facial nucleus and hypoglossal nucleus, the two nuclei that were under study in Aim 1 of this proposal. The facial nucleus contributes to the movements of at least seven different muscle groups in the head and neck, all of which have differential distributions within the somatotopic map of both the facial nucleus and orofacial motor cortex, while the hypoglossal nucleus contributes to the movements of ten different tongue regions (Cunningham et al., 2013; Afalo et al., 2006; Strick et al., 2021; Farina et al., 2023; Krammer et al., 1979; Sokoloff & Deacon, 1992). In Aim 1, we were interested in microstructural changes within these two nuclei associated with control of orofacial movements. In Aim 2, we are interested in whether structural connectivity between motor cortical areas that project to these nuclei within their brainstem level circuits and whether they have adapted in response to selection for sociocommunicative behaviors. We anticipate that macrostructural differences are likely to underlie behavioral differences in the Russian farm-foxes.

In primates and humans, orofacial movements are controlled by two descending parallel pathways which consist of descending projections from the primary motor cortex and additionally a parallel motor pathways descending from the rostral cingulate cortex which is considered to be part of the limbic system (Belyk et al., 2021a; Belyk et al., 2021b; Cerkevich et al., 2022; Zhang & Ghazanfar, 2022). Both of these regions contain body maps of the motor homunculus within each cytoarchitectonic region respectively, as do all premotor regions within the frontal cortex (Strick & Preston, 1978; Strick et al, 1998; Strick et al, 2021). The facial nucleus and hypoglossal nucleus additionally are the targets of long range projections from other premotor regions and these consist of either bilateral or unilateral input from areas like the supplementary motor region, the caudate region of the cingulate cortex, or the lateral and dorsal subregions of the premotor cortex (Müri, 2016; Morecraft et al., 2005; Morecraft et al., 2014).

In domesticated cats, a species more closely related to foxes than primates, there are three orofacial representations in the neocortex: the anterior part of the coronal gyrus (area C), the lateral wall of the presylvian sulcus (area P), and the ventral bank of the cruciate sulcus (area M) (Hiraba et al., 1993). These regions are largely thought to consist of cortical areas that control orofacial movements in species like the domesticated cat and dog. Both the facial nucleus and hypoglossal nucleus receive bilateral input from area C and area P, however the facial nucleus additionally receives a unilateral projection from area M while the hypoglossal nucleus receives additional input from the prorean gyrus (Schmitt & Gacek, 1986; Felleman et al., 1983; Hiraba et al., 2013; Hiraba et al., 2004; Hiraba et al., 2005). Coordination for orofacial movements consists of interspersed regions of fibers that both contribute to unilateral and bilateral control of the muscles for facial expression and movements of the tongue. Additionally, interhemispheric connections also seem to play a role in coordination of these muscles for orofacial movements (Hiraba et al., 2013).

Within canids, which include domesticated dogs and foxes, there are major differences between species-typical orofacial movements that involve movements controlled by the facial nucleus and hypoglossal nucleus (Fox, 1970; Sexton et al., 2023). While canids like coyotes and wolves have more complex facial movements, foxes have a greater range of orofacial mobility or range of motion in the orofacial regions (Fox, 1970). Even in closely related species that have similar orofacial behaviors within similar contexts can have varying degrees of expressiveness and reliance on specific orofacial movements for strong gestures; these repertoires are influenced by species-specific socialization processes or have changed during adaptive radiations as social structures have changed across canid species (Fox, 1970). Even though African wild dogs are fairly social organisms, they lack a large degree of facial mobility and thus primarily rely on vocalizations, upper body movements, and posture for communicative purposes (Robbins, 2000). Additionally, New Guinea singing dogs, anecdotally, are known for specific tongue postures where the tongue protrudes from the mouth during singing (Koler-Matznick et al., 2005).

Structural differences in the central nervous system across species likely contribute to behavioral variations, including motor behaviors, and may account for individual differences within a species. The importance of central motor cortical regions in controlling movement varies phylogenetically: in humans, damage to these areas results in significant motor impairment (Denny-Brown, 1950), while in non-human primates, damage causes transient impairments (Travis, 1955). In non-primate mammals, however, such damage typically affects fine motor control but does not extensively disrupt existing motor behaviors (Kawai et al., 2015; Lopes et al., 2023). This suggests that motor behaviors are governed by both cortical regions and brainstem-level circuits, with brainstem pathways helping to preserve behaviors when cortical regions are compromised.

In the context of this experiment, understanding the cortical input to nuclei controlling orofacial movement could shed light on how these movements are influenced by regions less conventionally associated with orofacial control. By comparing the robustness of these pathways and the connectivity patterns of these nuclei across experimental groups, we can gain insights into how selection for sociocommunicative behaviors might impact motor circuits for orofacial movement. For instance, in tame farm-foxes, which exhibit unique vocalizations directed at human handlers, the control of orofacial muscles for articulation may require more robust motor pathways, similar to the voluntary movement connectivity patterns seen in humans. Though facial expression in these foxes has not been fully documented, it is likely important for communication with human handlers, given human preferences for interpreting faces. We initially constructed a principal component analysis, to understand how orofacial and postural movements differ across groups. Our preliminary results show that tame foxes have orofacial and postural behaviors that diverge from both aggressive and conventional foxes. Theses results are shown in Figure XI. Data were provided kindly by Dr. Anna Kukekova from and we originally published in Kukekove et al, 2008.

One way of assessing robusticity and connectivity patterns underlying orofacial movements is through neuroimaging techniques like magnetic resonance imaging (MRI) combined with diffusion tensor imaging (DTI) which allows us to observe some of the structural organization of the central nervous system both in vivo and ex vivo. MRI offers high-resolution images of neural anatomical structures, while DTI is utilized for mapping the orientation and integrity of white matter tracts, which permit the reconstruction of white matter tracts that functionally and structurally connect different brain regions. Understanding the structural connectivity of cortical regions to the facial nucleus and hypoglossal nucleus, will provide some insight into the cortical control of the muscles that control facial expression and vocalization and to understand how widely this network is distributed within the cortex of these experimental lines. We anticipate that these networks will differ across the experimental and control lines, with the tame foxes exhibiting more robust pathways from frontal motor regions.

Aim 2 Materials and Methods:
The specimens described in Aim 1 underwent neuroimaging for this portion of the project. For imaging, the left hemisphere was placed in a waterproof container filled with polyethylene beads for stabilization and submerged in Fluorinert FC-770, a fluorocarbon that produces no MRI signal. Scanning was conducted on a 9.4 T Bruker magnet using a 7.2-cm-diameter RF coil and a RARE T2 sequence (2 averages, 13 ms TE, 2500 ms TR, rare factor 8), with a resolution of 300 μm^3 and a matrix size of $256 \times 100 \times 88$. This imaging protocol was adapted from Hecht et al. 2021, and the data were reanalyzed for this study.

Seed regions were placed at the facial and hypoglossal nuclei within the pontomedullary brainstem, based on cross-referencing with neuroanatomical atlases and methods from Zhang et al. 2020 for visualizing small brainstem structures. Image processing followed the pipeline from Hecht et al. (2015), using FSL software. T1-weighted images were denoised with SUSAN (Smith and Brady, 1997) and bias-corrected with FAST (Zhang et al., 2001), while DTI data were corrected for eddy current distortions with EDDY. Both T1 and B0 images underwent brain extraction with BET (Smith, 2002). BEDPOSTX (Behrens et al., 2003) modeled diffusion and estimated fiber crossings. Linear registration to T1 images was performed with FLIRT (Jenkinson and Smith, 2001), and nonlinear registration to a fox template was done with FNIRT (Andersson et al., 2007). PROBTRACKX (Behrens et al., 2003) was used for probabilistic tractography to map fibers through crossing regions and into gray matter.

Aim 2 Preliminary Data:
The facial nucleus and hypoglossal nucleus were identified for seeding without the MRI images from these scanned sections.

Aim 2 Significance and Interpretation:
If there are positive results, we might be able to interpret that selection for sociociommunicative behaviors can lead to increased cortical connectivity and thus circuit strengthening. Two ways in which we might observe circuits strengthening are increased number of axons or thickened axons along these interconnected regions. Increased number of axons might indicate that the number of neurons connecting these regions has increased and is most likely an adaptive byproduct that is more developmentally and genetically constrained. Thickening of myelin might indicate that increased cortical connectivity is due to adaptations that are a result of plasticity, and would be more likely to occur during the life of the organism, and not due primarily to genetic constraints. These would imply that an emphasis on social communication can lead to adaptive changes to motor circuits for social communication by different mechanisms.

If there are negative results, we might be able to interpret that selection for sociocommunicative behaviors doesn't require increased cortical input for adaptive change in social communication. This would mean that increased cortical input isn't the sole driver of changes to communicative modalities, taken with the findings from Aim 1, might support further investigation of cortical convergence on the brainstem level circuits for orofacial movements and additional investigation of the distribution of cell type within these circuits.

Because it is thought that the behaviors of these foxes are largely genetically constrained, but that there is both human-interaction and conspecific social interactions, that there will be contributions of both plasticity and developmental constraint in the results of this analysis. Because tame foxes have both species-specific behaviors, but additionally demonstrated an expanded behavioral repertoire that includes human interaction, we anticipate that these foxes might demonstrate more robust connectivity to the facial and hypoglossal nucleus. It is possible that both pathways would not be equally affected, if this is the case than we might be able to infer that human selection on affiliative behavior emphasizes one communicative modality over the other: Visual or auditory.

Summary and Future Directions:

These series of experiments were developed in order to clarify whether the effects of selection on social behaviors like facial expression and vocal communication can lead to evolved change in orofacial circuits that control important aspects of social communication leading to reorganization of cortical and brainstem level structures. Because social communication has played such an extensive role in the evolution of modern behaviors in humans and nonhuman animals, understanding selection for sociocommunicative behaviors may allow us some insight into phylogenetic patterns of brain organization that we see across mammalian species and the key features of human brain structures acquired over our long evolution. Our neuropil analysis will allows us to understand any microstructural changes to brainstem level nuclei, and our DTI analysis would allow us to further disambiguate if these circuits have been strengthened in the experimental foxes. Additionally, we would be able to determine that if circuit strengthening is mediated through an adaptation constrained by genetics, or if these changes are governed by plasticity within the lifetime of the organism. Future studies should highlight the need for understanding convergence of cortical and subcortical systems with brainstem level circuits to further elucidate structural differences that underlie functional differences observed in different types of stereotyped and nuanced behaviors. In an exploratory analysis, we found that both tame and aggressive foxes exhibit communicative behaviors that deviate from distributions found in natural communication systems, and that conventional foxes not under selection more closely resemble naturalistic power laws for communicative frequencies as described in Zipf's distribution. We hope to further explore the differences in communicative structures across these groups.

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TITLE: Evolved change in orofacial motor circuits under selection

Appendix with Figures:

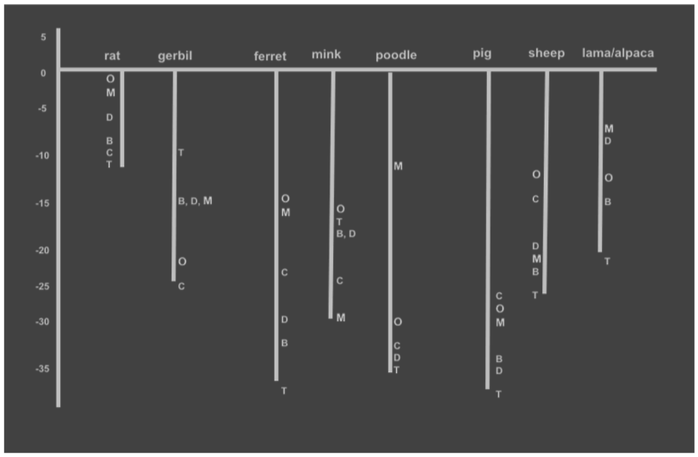

Figure I. Decrease in scaling size values of total brain and fundamental brain parts for several species from the wild to the domesticated form. In red, areas of the brainstem that contain the facial motor nucleus and hypoglossal nucleus. Adapted from Balcarcel et al., 2021. B: Brain weight, T: Telencephalon, D: Diencephalon, M= Mesencephalon, C: Cerebellum, O: Medulla oblongata. Conservation of the size of the mesencephalon and medulla oblongata are noted, especially in cases of a significant reduction in the domesticated species in volume of the telencephalon. The medulla oblongata and mesencephalon are the most reduced in gerbil, partially reduced in the poodle and pig, and less reduced in the rat, ferret, mink, sheep, and llama/alpaca. There are varying degrees of reduction in the medulla oblongata and mesencephalon of the members of carnivora represented in this figure suggesting that social structure might contribute overall to how conserved brainstem volume is in this order.

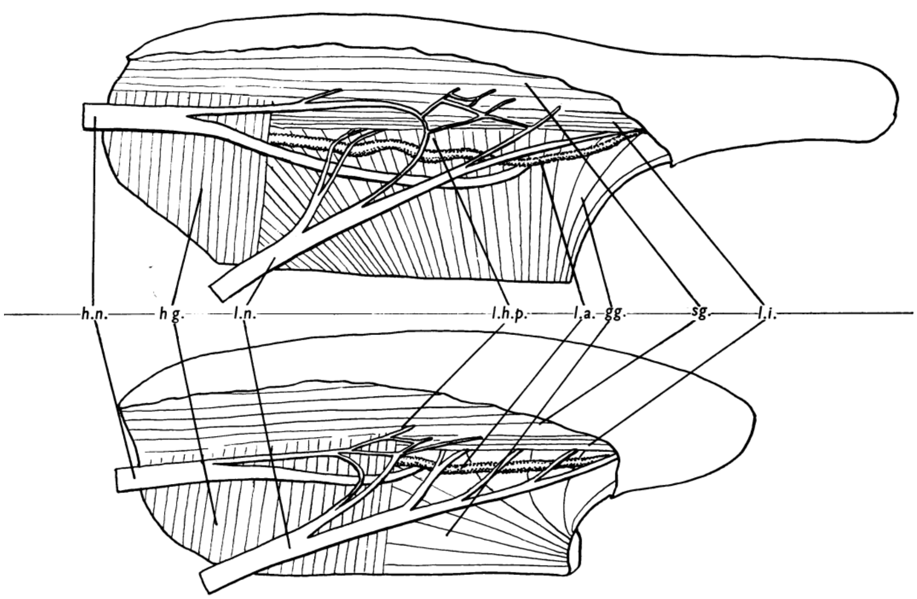

Figure II. Lingual and hypoglossal plexus in canines and humans. The human tongue has a forward extension of the hyoglossus muscle, and significant branches of the hypoglossal nerve that serve the posterior tongue musculature wrap around the front edge of this muscle. Dogs and cats have a simpler nerve plexus, formed when the lateral division of the hypoglossal nerve joins with one or two branches of the lingual nerve, with all resulting nerve bundles entering the styloglossus muscle (Fitzgerald & Law, 1958). Abbr. gg., genioglossus; hg., hyoglossus; h.n., hypoglossal nerve; L.a., lingual artery; l.h.p., lateral lingual-hypoglossal plexus; l.i., longitudinalis inferior; I.n., lingual nerve; sg., styloglossus.

J.M. Michlich

TITLE: Evolved change in orofacial motor circuits under selection

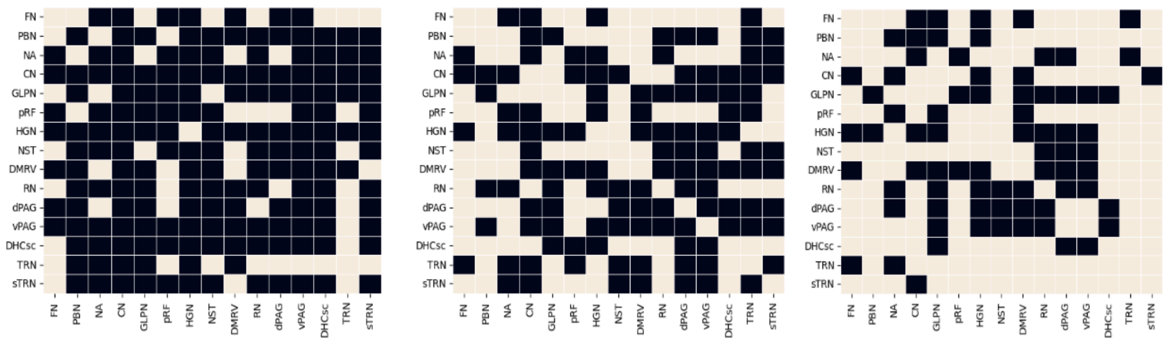

Figure IV. Brainstem connectivity in the macaque, rat, and cat for nuclei associated with vocal motor behaviors and facial expression. Neuroanatomical tract tracing studies are summarized in these plots of brainstem connectivity primarily within the brainstem to nuclei associated with vocal motor control and the control of facial expression. In order, macaque (left), cat (middle), rat (right). As brains scale in size, the interconnectedness of brainstem circuits decrease. For macaques, yellow squares indicate a connection and black squares indicate no connection. In rats, yellow squares indicate a connection and black squares indicate no connection. In the cat, yellow squares indicate a connection and black squares indicate no connection. Connectivity matrix was developed from information in Warren & May, 2013; Morecraft et al., 2014; Thomas & Jürgens, 1987; Rhoton, 1968; Miller & Strominger, 1973; Satoda et al., 1996; Vanderhost et al., 2000a; Vanderhost et al., 2000b; Pritchard et al., 2000; Beckstead et al., 1980; Mantyh, 1983; Hamilton at al., 1987; Qi & Kaas, 2006; Cheema et al., 1985; Goilli et al., 2001; Hannig & Jürgens, 200; Morecraft et al., 2014; Hardy & Leichnetz, 1981; Mantyh, 1981; von Monakow et al., 1979; Müller-Preuss & Jürgens, 1976; Larson et al., 1988; Saraswathi, 2003; Karim et al., 1981; Hinrichsen & Watson, 1983; Herbert et al., 1990; Cechetoo et al., 1985; Krout et al., 1998; Sawchenko, 1983; Núñez-Abades et al., 1990; Ennis & Rizvi, 1997; Atoji et al., 2005; Hayakawa et al., 1999; Weinberg & Rustioni, 1989; Berjemo et al., 2003; Contretas et al., 1982; Marfurt & Rajchert, 1991; Norgen, 1978; Hamilton & Norgren, 1984; Housley et al., 1987; Fay & Norgren, 1997; Ter Horst et al., 1991; Yasui et al., 2001; Yasui et al., 1998; Leong et al., 1984; Beitz, 1982; Marchand & Hagino, 1983; Godefroy et al., 1998; Pinganaud et al., 1999.

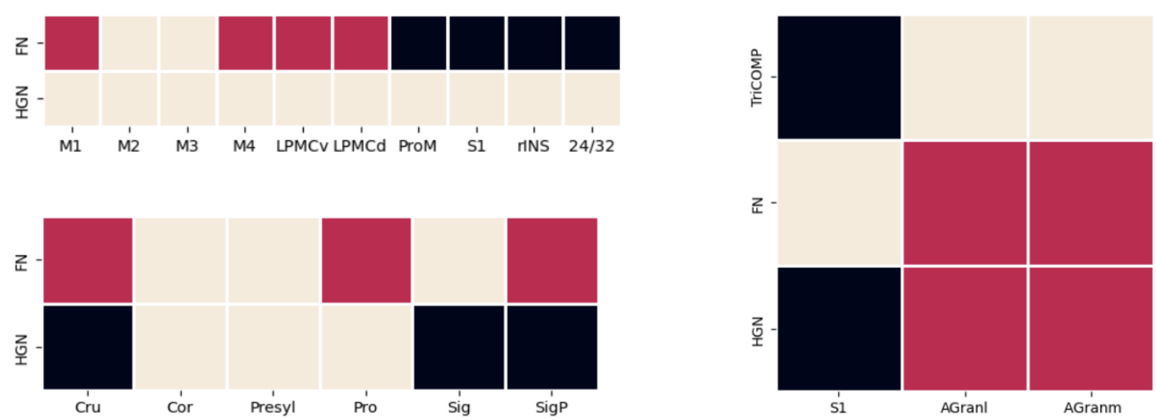

Figure V. Cortical connectivity in the macaque, rat/mouse, and cat for the facial nucleus and hypoglossal nucleus. Taken with the brainstem connectivity matrix, cortical connectivity increases when brains scale to a larger size and brainstem circuits are less interconnected. Macaque (top left), rat* (right), and cat (bottom left). Red squares indicate unilateral connections; yellow squares indicate bilateral connections, and black squares represent no connection. *Mouse and rat data were combined minimally when similar patterns could be assumed across species. Also important to note that in rodents, jaw, orofacial, and forelimb movements are largely influenced by the motor trigeminal complex and not the facial nucleus and hypoglossal nucleus. Connectivity matrix was developed from information in Morecraft et al., 2001; Morecraft et al., 2014; Schmitt & Gacek, 1986; Hiraba et al., 2013; Hiraba & Sato, 2004; Hiraba et al., 2005; Felleman et al., 1983; Matano et al., 1972; Jeong et al., 2016; Takatoh et al., 2021; Petrof et al, 2015.

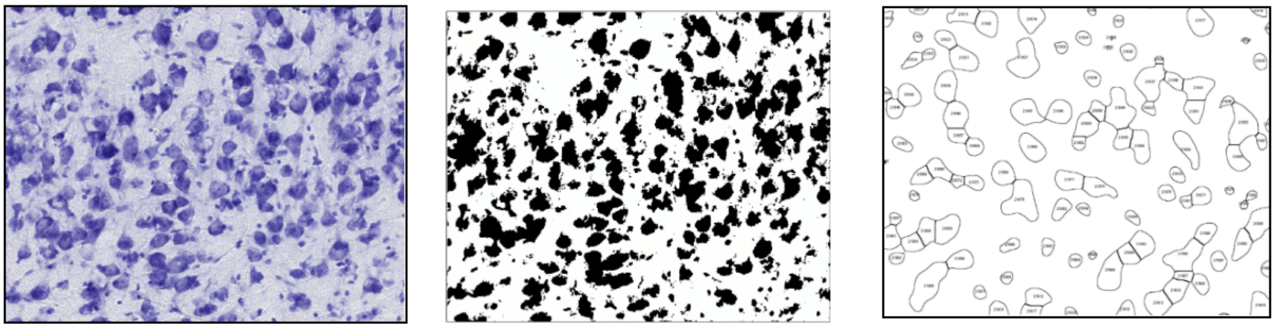

Figure VI. Images are imported and scaled in the open-source software ImageJ. For the neuropil pipeline, images are converted to binary images to register the percentage of the ROI consisting of cell bodies versus neuropil (white matter). In the cell profile pipeline, a mask is applied to areas where there are cell bodies, and the area is calculated after the image is converted from pixels to microns. Additionally, cell diameter can be derived with a similar mask.

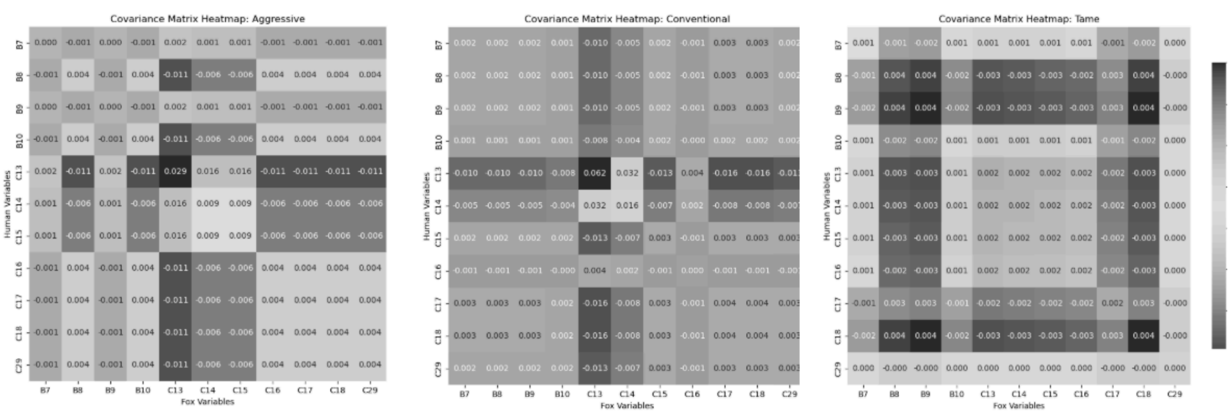

Figure VII. Behavioral covariance of approach behaviors between farm-foxes and human handlers in which handlers initiated physical contact. Tame foxes exhibit positive responses to human handlers that allows for extended contact, while conventional and aggressive foxes exhibit a pattern of inverse covariance when presented with human contact.

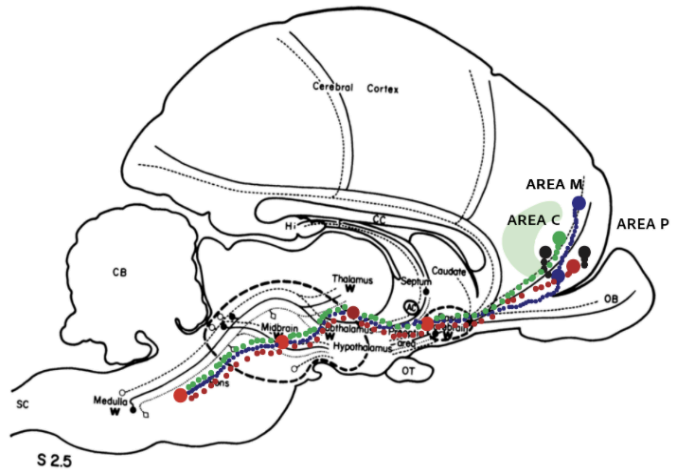

Figure VIII. Circuit for orofacial motor control in domesticated cats. Motor areas M and P contribute to the cortical control of orofacial movements in domesticated cats, and likely in the farm-foxes. This circuit descends through the basal ganglia and motor thalamus, to the facial nucleus and hypoglossal nucleus in the pontomedullary brainstem. Both Area M and Area P both have commissural connections that allow for coordination of these motor nuclei across motor hemispheres. Unilateral and bilateral control of these nuclei are interspersed within these motor regions across both hemispheres. Area C and the surrounding somatosensory cortex also project unilaterally to the facial and hypoglossal nucleus. Image adapted from Zeman, 2001.

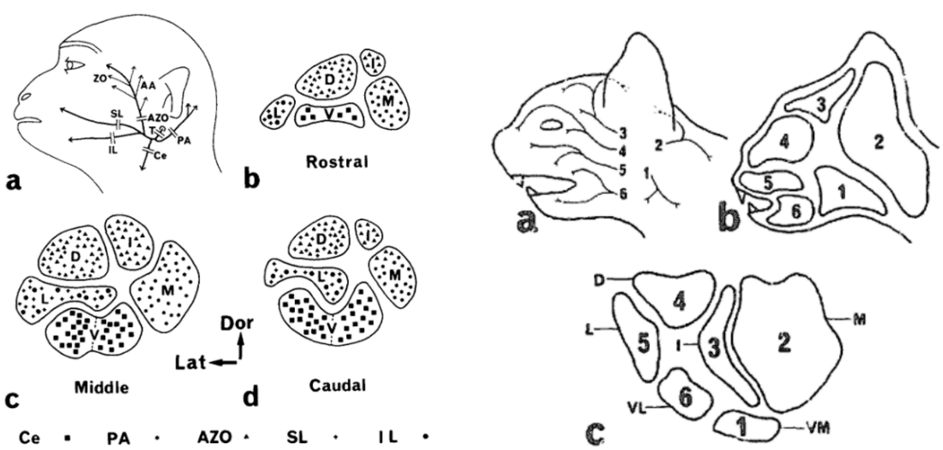

Figure X. Distribution of innervation patterns of the facial nucleus with respect to the main branches of the facial nerve in the macaque and domesticated cat. From Satoda et al., 1987 and Kume et al., 1978.. For the macaque: The cervical branch primarily originates from the ventro and ventrolateral facial nucleus. The posterior auricle branch primarily originates from the middle and caudal medio-lateral subdivisions. The anterior auricular-zygomatico-orbital branch originates from the dorsal and intermediate subdivisions. The superior labial branch originates from the lateral subdivision and the inferior labial branch from the lateral subdivisions. For the cat: The cervical branch originates from the ventromedial subdivision. The posterior auricular branch originates from the medial subdivision. The temporal branch originates from the intermediate subdivision. The zygomatico-orbital branch originates from the dorsal subdivision. The superior labial branch originates from the lateral subdivision. The inferior labial branch originates in the ventrolateral subdivision.

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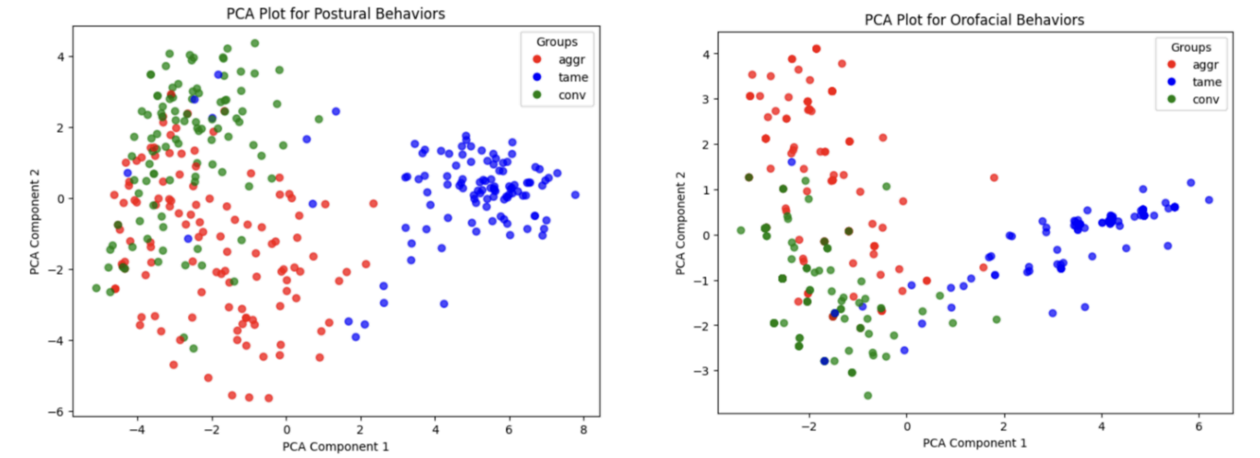

Figure XI. PCA of orofacial and postural behaviors in tame, aggressive, and conventional foxes. Tame foxes show a marked difference in postural and orofacial behaviors from aggressive and conventional foxes. Aggressive foxes and conventional foxes show more distinct clusters in postural behaviors than in orofacial behaviors, where there clusters overlap significantly. The orofacial and postural behaviors of tame foxes are the most isolated from the behaviors of the other two groups.

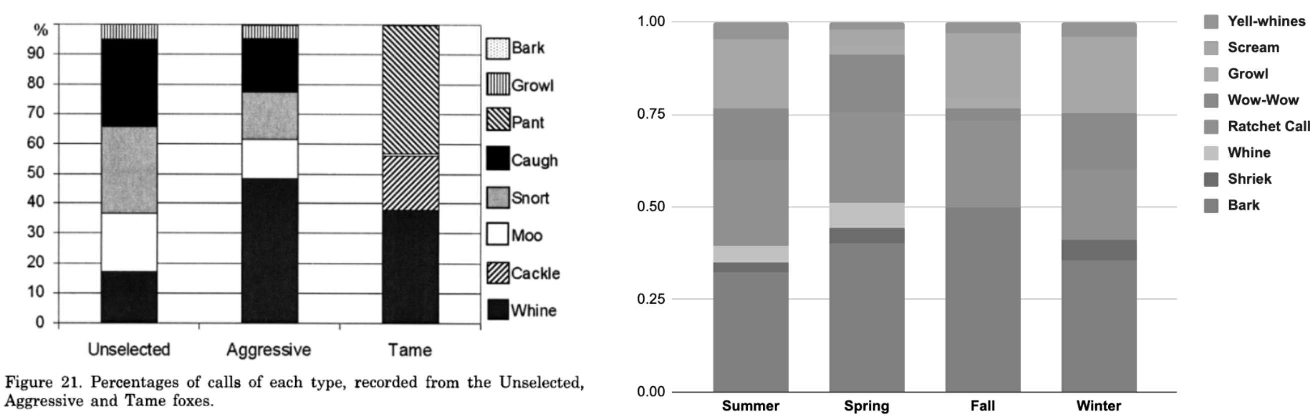

Figure XIII. Differences in vocalization in the farm-foxes versus seasonal fluctuations in vocal behaviors in wild-caught foxes. Aggressive and conventional foxes have variability in vocal signals more similar to wild-caught foxes, while tame farm-foxes have a reduction in the vocal repertoire. Tame foxes mostly Vocalize with whines, cackles, and pants. Whines in wild-caught foxes typically are only present during the reproductive season. It is notable that tame farm-foxes have estrous twice a year.

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